

Week 01 Task 1

Cyber Security Intro + What is a SOC?

CandidateName: Asad Ali
Reference Number: TBS-INT-2026-445
Submission Date: 16|10|2026
Submitted To: TechBiz Security

Week 01: Cyber security Basics CIA triad

01: The CIA Triad: CIA stands for **confedentiality**, **Integrity** and **Availability**. Complete cyber security is based on 3 principles these are developed in 1980 so that every non technical person also understands the cyber security in a simplified way.

1) **Confidentiality**: This ensures that information is accessed only by authorized people.

Example: No one should be able to access someone's personal mobile phone

2) **Integrity**: Data remains same, accurate unaltered from sending end to receiving end.

Example: User A transfer email to user B should receive email that user A sends without any changing

3) **Availability**: Ensures that system and information remains available for authorized users when needed.

Example: Websites, apps, and servers should be up and running 24/7.

02: Threat vs Risk vs Vulnerability

Threat: A potential event, actor, or action that can exploit a vulnerability and cause harm to a system, network, or organization.

Risk: The potential for loss when a threat exploits a vulnerability, determined by considering the likelihood of occurrence and the potential impact.

Vulnerability: Software deployment misconfigurations and insecure coding practices can introduce security weaknesses that attackers may exploit, potentially leading to unauthorized access or compromise of the affected device or system.

03: Definition of SOC

A Security Operations Center (SOC) is a centralized team or department within an organization responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity threats and security incidents across the organization's IT environment.

04: Purpose of SOC

The primary purpose of a SOC is to protect an organization's systems, networks, applications, and data from cyber threats. It continuously monitors security events, identifies suspicious or malicious activities, investigates potential incidents, and coordinates an appropriate response to minimize their impact.

05: Responsibilities of L1 L2 , L3 SOC analyst

L1 Responsibility:

SOC L1 Analyst is a person who is the first line of defence to SOC department. Key responsibilities are mentioned below.

- **Security Monitoring:** Monitor security alerts and events generated by SIEM, EDR, IDS/IPS, firewalls, and other security tools.
- **Alert Triage:** Review and prioritize alerts based on their severity, potential impact, and indicators of compromise.
- **Initial Investigation:** Analyze logs, IP addresses, domains, URLs, file hashes, authentication events, and network activity to determine whether an alert is legitimate or a false positive.
- **Incident Classification:** Categorize security incidents such as phishing, malware, brute-force attempts, suspicious logins, and unauthorized access.
- **Incident Escalation:** Escalate confirmed or high-severity incidents to L2/L3 analysts according to established procedures.
- **Following Playbooks:** Follow established SOC procedures and incident-response playbooks when investigating and handling alerts.

L2 Analyst Responsibility/ Incident Responder:

SOC L2 Analyst performs deeper investigation on alerts escalated by L1 Analyst. L2 analysts validate threats, determine their scope and impact, and perform incident-response activities to contain and remediate confirmed security incidents.

- **Advanced Investigation:** Perform detailed analysis of escalated alerts using SIEM, EDR, IDS/IPS, firewall, and other security data.
- **Incident Response:** Respond to confirmed security incidents by following established incident-response procedures and playbooks.
- **Threat Analysis:** Analyze indicators of compromise (IOCs), suspicious IP addresses, domains, URLs, file hashes, processes, and network activity.
- **Containment:** Take or coordinate actions such as isolating compromised endpoints, blocking malicious IP addresses or domains, and disabling compromised accounts.
- **Root Cause Analysis:** Investigate how an incident occurred, identify the initial attack vector, and determine which systems or accounts were affected.
- **Remediation and Recovery:** Assist in removing malicious artifacts, addressing security weaknesses, restoring affected systems, and verifying that the threat has been eliminated.
- **Incident Escalation:** Escalate highly complex incidents, advanced threats, or cases requiring specialized expertise to L3 analysts, threat hunters, DFIR teams, or other relevant teams.
- **Documentation and Reporting:** Document investigation findings, evidence, response actions, impact, and recommendations for future prevention.

L3 Analyst Responsibility/ Threat Hunter:

A SOC Level 3 (L3) Analyst is a senior security professional responsible for investigating complex cyber threats, performing proactive threat hunting, and supporting the response to sophisticated security incidents. L3 analysts also help improve the organization's overall threat detection and response capabilities.

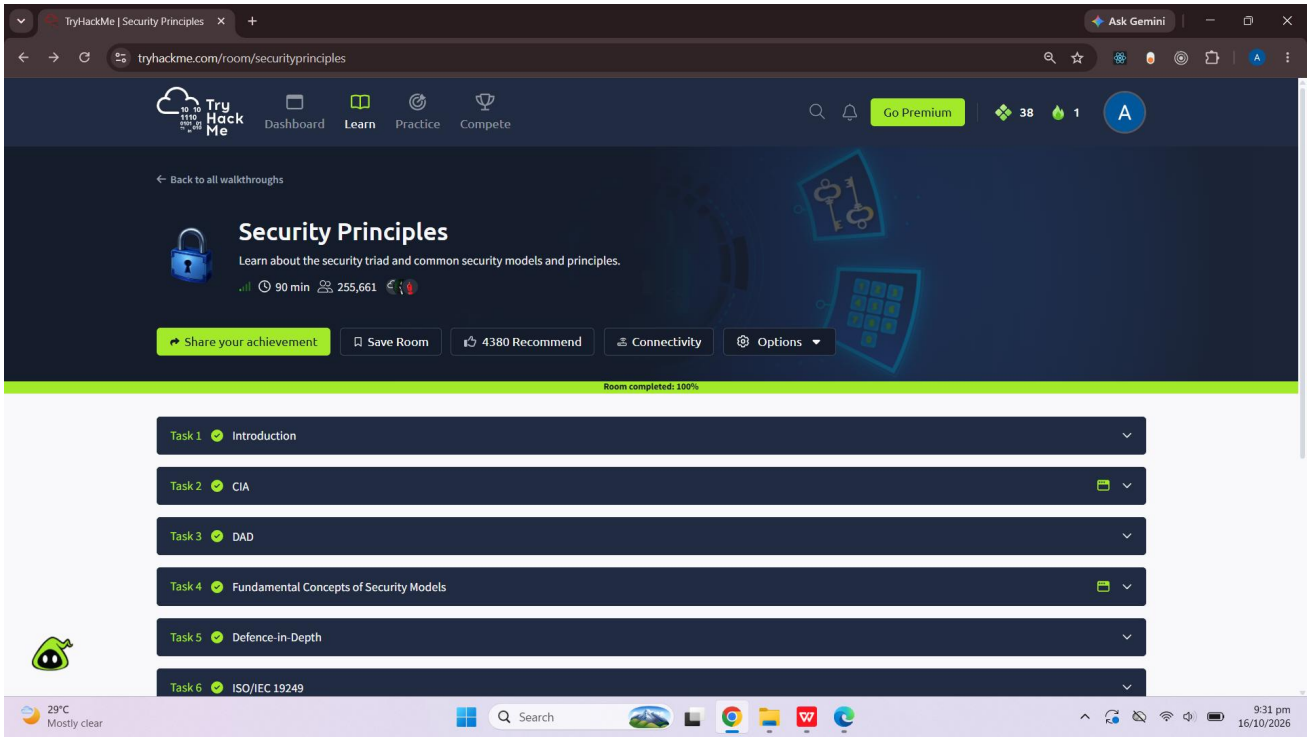
Key responsibilities include:

- **Threat Hunting:** Proactively search across networks, endpoints, and logs for hidden threats that may not have triggered existing security alerts.
- **Advanced Incident Investigation:** Investigate complex or high-severity incidents escalated by L2 analysts.
- **Advanced Threat Analysis:** Analyze sophisticated attacks, attacker behavior, Indicators of Compromise (IOCs), and Tactics, Techniques, and Procedures (TTPs).
- **Detection Engineering:** Develop and improve SIEM detection rules, correlation rules, queries, and alerting mechanisms to identify emerging threats.
- **Threat Intelligence:** Use threat intelligence to investigate emerging attack techniques, malicious infrastructure, and threat actors relevant to the organization.
- **Malware Analysis:** Perform or support advanced analysis of suspicious files and malware to understand their behavior and impact.
- **Root Cause Analysis:** Determine the attack path, initial entry point, affected assets, and underlying security weaknesses behind complex incidents.
- **SOC Improvement:** Provide recommendations to improve security controls, detection capabilities, playbooks, and incident-response processes.
- **L1/L2 Support:** Provide technical guidance to junior analysts and assist with complex investigations and escalations.

06: Difference between Red Team and Blue team

Red Team	Blue Team
Simulates real-world attackers to identify and exploit security weaknesses.	Defends the organization by detecting, investigating, and responding to security threats.
Performs penetration testing, vulnerability exploitation, social engineering, and adversary simulation.	Performs security monitoring, threat detection, incident response, threat hunting, and security analysis.
Takes an **offensive/proactive** approach to test how effectively the organization can withstand attacks.	Takes a **defensive/reactive and proactive** approach to detect, contain, and prevent attacks.

07: TryHackMe Security principles Room



08: TryHackMe Pentesting Fundamentals Room

